DEVELOPMENT OF AN LL(1) PARSER

WITH FIRST, FOLLOW AND PARSING TABLE GENERATION

ABSTRACT

Syntax analysis is an important phase of a compiler in which the grammatical structure of source code is checked according to a set of production rules. This project focuses on the development of an LL(1) parser that automatically computes FIRST and FOLLOW sets and constructs an LL(1) predictive parsing table for a given context-free grammar. The system takes grammar productions as input, identifies terminals and non-terminals, and processes the productions to calculate their FIRST and FOLLOW sets. Using these sets, it generates a parsing table that maps each non-terminal and input symbol to the appropriate production. The parser then uses a stack and an input buffer to analyze a given string, selecting productions from the parsing table and determining whether the string is accepted or rejected.

The proposed system helps users understand the internal working of predictive parsing by presenting the intermediate results, including FIRST sets, FOLLOW sets, and the parsing table. It also checks whether the grammar satisfies the LL(1) condition by identifying conflicts such as multiple productions being assigned to the same table entry. By automating these steps, the project reduces manual calculation errors and provides a clear, practical way to study top-down parsing in compiler design. The implementation demonstrates how grammar analysis and table-driven parsing work together to validate input strings efficiently.

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